

ex2 imagine a face recog system that identifies a person:-

- input layer :- image pixels (eyes nose mouth).
- Hidden layer :- learns how these features are connected.
- Output layer :- predicts the persons name.

• Advantages it can handle complex problems like speech recognition and handwriting detection.

③ Feed forward Neural Network

it moves data in one direction (from input to output) without loops.

~~ex~~ simple and fast, but can not "remember" previous inputs (like a chatbot remembering past messages)

④ Backpropagation Neural Network. Is a feed back loop that helps the team of helpers get better at recognizing animals.

Repeat and improve.

The team keeps and practicing with different pictures making ~~poor~~ guesses, checking answers, and adjusting their decision process.

⇒ What is agents environments, it refers to the external world that the agent interacts with.

Here are the properties of an environment

1. Fully observable.

The agent has complete knowledge of the environment.

ex: A chess game: the agent (player) can see the entire board.

2. Partially observable.

The agent has limited knowledge of the environment.

ex: A poker game: the agent (player) can only see their cards.

3. Deterministic: The environment's response to the agent's actions is predictable.

4. Static: The environment does not change.

5. Dynamic: The env changes over time.

These properties help us understand and design agents that can effectively interact with their environments.

⇒ Types of agents

1) Simple Reflex Agents: Only reacts to current situation.

ex: thermostat: turns heating/cooling on/off based on current temperature.

2) Model Based Agents: Uses internal model to predict future situations.

ex:

weather forecasting app: predicts future weather based on current data and models.

3) Goal-Based Agents: work towards achieving specific goals.

ex:

GPS: navigation: find best route to destination (Goal)

4) Utility-Based Agents: Makes decisions based on maximizing utility (value)

ex:

Recommendation system: suggests products based on users past behavior and preferences. (Utility)

5) Learning Agents: improves performance over time through learning.

ex: personal assistant (eg Siri, Alexa)

learns users habits and preferences over time.

⇒ Agents:

1. it perceives its environment (eg sensors, data)
2. Take actions to achieve a goal (eg move, decide)
3. Learns from experience to improve performance.

→ Think of an agent like a robot that:

- sees the world around it (perception)
 - Decides what to do (action)
 - Gets better with practice (learning)
- ex: self driving car. An agent that:
- perceives the road environment (cameras, sensors)
 - Take actions to navigate (steering, acceleration)
 - Learns to improve driving skills through experience

→ each type of agent has its own strengths and weaknesses, and is suited for specific applications -

agent consist on:

1. takes initiative
2. Self-governing
3. learns and improves